# ■ Machine Learning Formula Handbook (Running PDF)

## ■■ Date: 10/12/2025

### *Naive Bayes Theorem:*

P(A|B) = (P(B|A) * P(A)) / P(B)

### *Gaussian Naive Bayes:*

$P(x|y) = (1 / \sqrt{(2\pi\sigma^2)}) * \exp(-(x - \mu)^2 / (2\sigma^2))$

### *Bernoulli Naive Bayes:*

Feature values are binary: 0 or 1

### *Multinomial Naive Bayes:*

Used for frequency/count-based features